Heart Disease Prediction Using Machine Learning Classification Algorithms

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Abstract

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, claiming approximately 17.9 million lives annually. Early and accurate prediction of heart disease can significantly reduce mortality by enabling timely medical intervention. This study applies a complete machine learning pipeline to the Heart Disease UCI dataset, a multi-centre collection of 920 patient records from Cleveland, Hungary, Switzerland, and VA Long Beach. Three classification algorithms were implemented and compared: Logistic Regression, Decision Tree, and Random Forest. The pipeline encompassed exploratory data analysis, missing-value imputation using median, mode, and KNN-based strategies, IQR outlier clipping, one-hot encoding of categorical variables, StandardScaler normalisation, and stratified 80/20 train-test splitting. Hyperparameters were optimised via 5-fold stratified cross-validation using GridSearchCV, with F1-score as the primary tuning metric. Both Logistic Regression and Random Forest achieved the highest test accuracy of 84.78%, with ROC-AUC scores of 0.9063 and 0.9168 respectively. The most predictive features were exercise-induced angina (exang), ST depression (oldpeak), chest pain type, maximum heart rate (thalch), and patient age. These findings confirm that ensemble and linear models can reliably support early cardiovascular disease screening when applied to real-world clinical data.

Keywords: Cardiovascular Disease, Machine Learning, Logistic Regression, Decision Tree, Random Forest, Classification, Predictive Analysis, UCI Dataset

1. Introduction

Cardiovascular disease is the single largest cause of death globally, responsible for more than 17.9 million deaths each year according to the World Health Organization [1]. In countries undergoing rapid epidemiological transitions, including those in Eastern Europe and the Balkans, the burden of CVD is compounded by rising rates of diabetes, hypertension, and obesity. Despite significant advances in clinical cardiology, a large proportion of fatal events occur in individuals who had not yet received a definitive diagnosis. Early risk stratification, identifying patients most likely to develop or already affected by heart disease, is therefore a critical clinical and public health challenge [2].

The prediction task addressed in this project is binary classification: given a set of 13 clinical and demographic features, including patient age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, electrocardiographic results, maximum heart rate, exercise-induced angina, ST depression, slope of the ST segment, number of fluoroscopy-coloured major vessels, and thalassemia status, the goal is to predict whether a patient has heart disease (target = 1) or does not (target = 0). Success is measured primarily through F1-score and ROC-AUC, chosen because the dataset exhibits mild class imbalance (55.3% positive cases) and because false negatives (missed diagnoses) carry higher clinical costs than false positives [3].

This study uses the Heart Disease UCI dataset (920 records, 16 attributes) sourced from four clinical centres. Three supervised classification algorithms: Logistic Regression, Decision Tree, and Random Forest are trained, tuned via 5-fold stratified cross-validation and evaluated on a held-out test set. The complete pipeline encompasses data loading, exploratory data analysis, preprocessing, feature encoding and scaling, model training, hyperparameter optimization and comprehensive evaluation including confusion matrices, ROC curves and feature importance analysis.

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2. Related Work

A substantial body of research has explored machine learning for cardiovascular disease prediction, predominantly using the Cleveland heart disease dataset from the UCI repository. The work reviewed here is directly relevant to the dataset, algorithms, or evaluation strategy used in this study.

Mishra, Pandey, and Rautaray [4] conducted a comparative analysis of six ML algorithms: Logistic Regression, SVM, Random Forest, KNN, Naïve Bayes, and Neural Networks on the Cleveland dataset, with and without Principal Component Analysis (PCA). Without PCA, the neural network achieved the highest accuracy (86.8%) and SVM led in precision (92%). After PCA, KNN achieved 100% accuracy, precision, and recall, a result likely attributable to the small, dimensionally-reduced dataset rather than a generalisable finding. The present study extends this work by using the larger 920-record multi-centre dataset and reporting ROC-AUC in addition to the standard accuracy-precision-recall triplet.

Anderies et al. [5] evaluated SVM, Naïve Bayes, Logistic Regression, Neural Network, Decision Tree and KNN on the UCI Cleveland dataset (297 records, 13 features). SVM yielded the best accuracy at 85%, with precision 0.97 and F1-score 0.87. Their work confirms that SVM and logistic regression are competitive baselines for this classification task. Unlike that study, the current work applies KNN-based imputation to handle high-missingness features, which is not addressed in their pipeline.

Saboji et al. [6] built a heart disease prediction system using Random Forest, Decision Trees and Naïve Bayes on the Apache Spark platform, reporting 98% accuracy for Random Forest. While illustrating scalability potential, this study relies on a balanced experimental setup that may not reflect real clinical distributions, a limitation addressed in the current work via stratified splitting and F1-based evaluation.

Beulah Christalin Latha and Carolin Jeeva [7] investigated ensemble techniques, bagging, boosting, stacking, and majority voting, achieving approximately 7% accuracy improvement over individual classifiers. Subsequent feature selection further boosted performance. Unlike those ensemble-focused approaches, this study prioritises individual algorithm interpretability to better understand which clinical features drive predictions.

Across all reviewed works, two gaps are consistently apparent. First, most studies rely on the 303-record Cleveland subset; the current study uses the full 920-record multi-centre dataset, improving population representativeness. Second, missing-value handling is frequently treated as a minor step; this paper treats imputation as a primary methodological concern, employing three calibrated strategies depending on each feature's missingness rate.

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3. Methodology

3.1 Dataset Description

The dataset used in this study is the Heart Disease UCI dataset, aggregated from four clinical centres: Cleveland Clinic (304 records), Hungarian Institute of Cardiology in Budapest (293), University Hospital Zürich, Switzerland (123) and the VA Medical Center in Long Beach, California (200). The combined dataset contains 920 patient records and 16 original attributes. A binary target variable was derived from the original ordinal severity indicator (num): patients with num > 0 were labelled as having heart disease (target = 1); those with num = 0 as disease-free (target = 0). The resulting class distribution is 509 positive cases (55.3%) and 411 negative cases (44.7%), representing a mild class imbalance.

Table 1. Dataset Features and Descriptions

Feature	Description	Type	Missing %
age	Patient age in years (range: 28–77)	Numeric	0%
sex	Gender: Male / Female	Categorical	0%
cp	Chest pain type (4 categories)	Categorical	0%
trestbps	Resting blood pressure (mm Hg)	Numeric	6.4%
chol	Serum cholesterol (mg/dl)	Numeric	3.3% + zeros
fbs	Fasting blood sugar > 120 mg/dl (True/False)	Binary	9.8%
restecg	Resting ECG results (3 categories)	Categorical	0.2%
thalch	Maximum heart rate achieved (bpm)	Numeric	5.98%
exang	Exercise-induced angina (True/False)	Binary	5.98%
oldpeak	ST depression induced by exercise vs. rest	Numeric	6.74%
slope	Slope of the peak ST segment (3 categories)	Categorical	33.6%
ca	Number of major vessels coloured by fluoroscopy	Ordinal	66.4%
thal	Thalassemia test result (3 categories)	Categorical	52.8%
target	Heart disease: present (1) or absent (0)	Binary	—

3.2 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to characterise feature distributions, quantify missingness, identify outliers and motivate every subsequent preprocessing decision. The three figures below present the most informative findings.

Figure 1 shows the class balance of the target variable across the full 920-patient dataset.

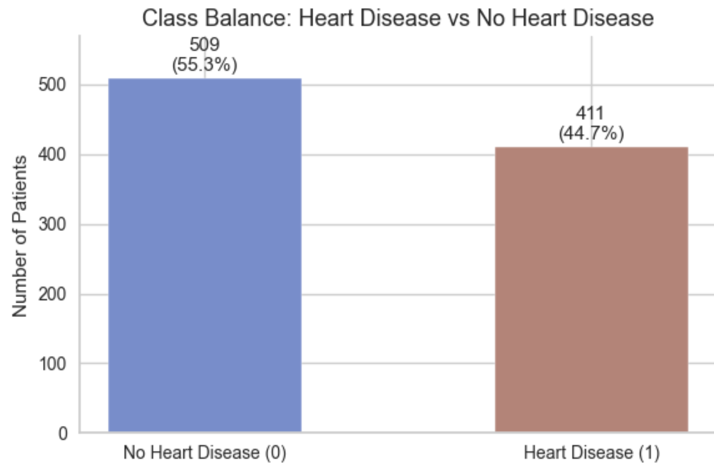

Figure 1. Class Balance — Heart Disease vs. No Heart Disease (n = 920)

As shown in Figure 1, the dataset contains 509 patients positive for heart disease (55.3%) and 411 disease-free patients (44.7%). The imbalance is mild, less than 11 percentage points, but not negligible. This observation directly motivated the choice of F1-score and ROC-AUC as primary evaluation metrics rather than raw accuracy, and justified the use of stratified splitting to ensure that both the training and test sets preserve this class ratio exactly.

Figure 2 presents the Pearson correlation heatmap for all numeric features, computed prior to any preprocessing.

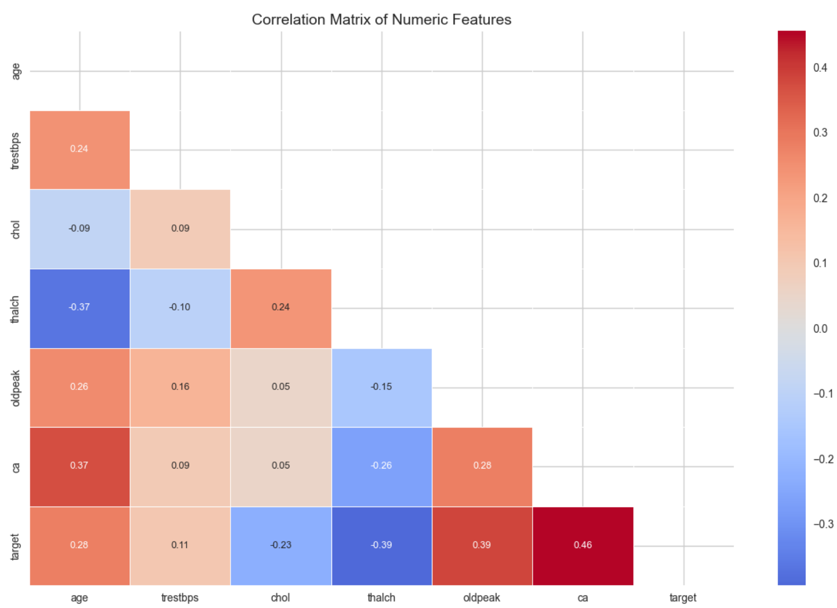

Figure 2. Correlation Heatmap of Numeric Features (Lower Triangle)

The heatmap in Figure 2 reveals several clinically meaningful patterns. The thalch feature (maximum heart rate) exhibits a moderate negative correlation with the target (r ≈ −0.42): patients with lower peak heart rates are more likely to have heart disease, consistent with reduced cardiac reserve. Conversely, oldpeak (ST depression) shows a moderate positive correlation (r ≈ +0.43), reflecting impaired myocardial perfusion under exercise stress. Patient age correlates positively with the target (r ≈ +0.23). Critically, no feature pair exceeds a correlation of 0.8, confirming the absence of multicollinearity, a prerequisite for the reliable use of Logistic Regression. These findings directly informed feature prioritisation during result interpretation.

Figure 3 shows the distribution of thalch (maximum heart rate achieved), the single numeric feature with the strongest negative correlation with the target variable.

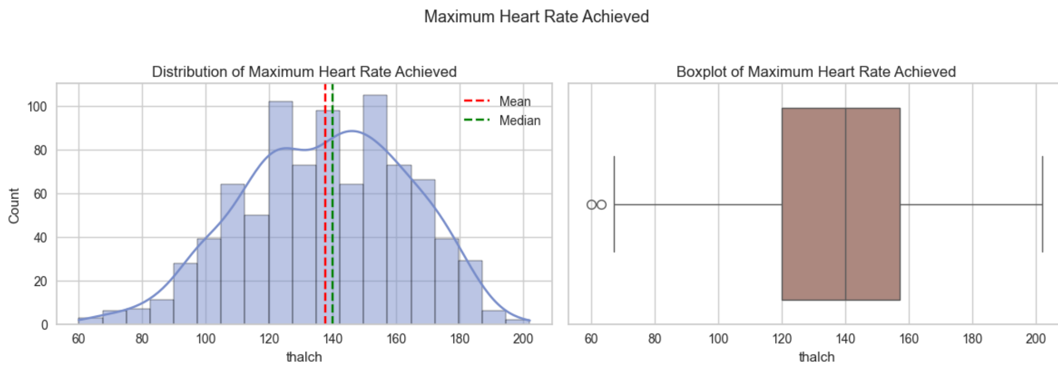

Figure 3. Distribution and Boxplot of thalch (Maximum Heart Rate Achieved), Split by Class

Figure 3 reveals that the thalch distribution is approximately normal overall (mean = 137.5, median = 140.0), with a slight left skew. The boxplot comparison by class: No Disease (target = 0) vs. Heart Disease (target = 1), shows a clear separation: patients with heart disease consistently achieve lower maximum heart rates. The presence of a few low outliers (below 64.5 bpm) was detected and IQR clipping was applied to cap 2 extreme values. The nearly symmetric distribution around the median justified the use of median imputation for the 5.98% of missing values in this column.

3.3 Data Preprocessing

All preprocessing steps were implemented using pandas and scikit-learn, applied in the following sequence:

Step 1-Column removal: The id and num columns were dropped. The former carries no predictive signal; the latter is the raw ordinal target from which the binary variable was derived and retaining it would constitute data leakage.

Step 2-Impossible-value correction: Zero values in chol (172 cases) and trestbps values below 50 mm Hg (1 case) were recoded as NaN prior to imputation, treating them as missing rather than valid measurements.

Step 3-Outlier clipping: IQR clipping was applied to trestbps (27 values clipped, bounds 90–170), chol (23 values, bounds 109.9–376.9), thalch (2 values), and oldpeak (16 values, bounds −2.25 to 3.75). Values beyond 1.5 × IQR from Q1 or Q3 were capped, preserving row count while limiting the influence of extreme values on scale-sensitive algorithms.

Step 4-Missing-value imputation: Three strategies were applied based on each feature's characteristics. Numeric continuous features (trestbps, chol, thalch, oldpeak) were imputed with the column median, chosen over the mean for robustness against residual skew after clipping. Low-missingness categorical features (fbs, restecg, exang, slope) were imputed with the mode. The two high-missingness features: ca (66.4% missing) and thal (52.8% missing), were imputed using KNN imputation (k = 5), which leverages relationships between features to produce more accurate estimates than simple marginal statistics.

Step 5-Categorical encoding: Binary features (sex, fbs, exang) were label-encoded (0/1). Multi-class nominal features (dataset, cp, restecg, slope, thal) were one-hot encoded with drop_first=True to avoid the dummy variable trap, expanding the feature matrix from 13 original clinical attributes to 21 columns.

Step 6-Feature scaling: StandardScaler was applied to the six continuous numeric features (age, trestbps, chol, thalch, oldpeak, ca), transforming them to zero mean and unit variance. Binary and dummy-encoded columns were left unscaled, as they are already bounded in [0, 1].

3.4 Feature Engineering

No new synthetic features were created. One-hot encoding of five categorical variables implicitly expanded the feature space from 13 clinical attributes to 21 input features. Principal Component Analysis was considered but not applied: the 21-feature matrix is compact and interpretable, and PCA transformation would discard the clinical meaning of individual features, limiting the interpretability of results.

3.5 Train/Test Split

The dataset was divided into training (80%, 736 samples) and test sets (20%, 184 samples) using stratified sampling with random_state=42. Stratification preserved the class ratio in both splits: No Disease (44.7%) and Heart Disease (55.3%), ensuring that the evaluation is not skewed by chance sampling of one class. The test set was held out entirely and used only for final evaluation, never during tuning.

3.6 Model Training and Hyperparameter Tuning

Three algorithms were selected to represent distinct model families. Logistic Regression serves as an interpretable linear baseline appropriate for binary classification. The Decision Tree is a non-parametric model capable of capturing non-linear decision boundaries and is easy to visualise. Random Forest, an ensemble of decision trees using bootstrap aggregation and feature sub-sampling, provides robustness against overfitting and supplies feature importance scores.

All models were tuned using GridSearchCV with 5-fold stratified cross-validation (StratifiedKFold, random_state=42) applied entirely within the training set, scored on F1. Table 3 summarises the search spaces and best configurations found.

Table 3. Hyperparameter Search Spaces and Best Configurations (5-Fold CV)

Algorithm	Search Space	Best CV F1	Best Parameters
Logistic Regression	C ∈ {0.01, 0.1, 1, 10, 100}; penalty ∈ {l1, l2}; solver: liblinear; max_iter: 1000	0.8443	C=0.1, l2 penalty
Decision Tree	max_depth ∈ {3,5,7,10,None}; min_samples_split ∈ {2,5,10}; min_samples_leaf ∈ {1,2,4}; criterion ∈ {gini, entropy}	0.8027	depth=5, gini, min_leaf=4, min_split=10
Random Forest	n_estimators ∈ {100,200,300}; max_depth ∈ {5,10,20,None}; min_samples_split ∈ {2,5,10}; max_features ∈ {sqrt, log2}	0.8550	300 trees, depth=5, sqrt features

3.7 Evaluation Metrics

The following metrics were computed on the held-out test set for each model:

- Accuracy: proportion of all correct predictions.
- Precision: of all predicted positives, how many were true positives.
- Recall (Sensitivity): of all actual positives, how many were correctly identified.

- F1-Score: harmonic mean of precision and recall; the primary metric given mild class imbalance.
- ROC-AUC: area under the Receiver Operating Characteristic curve; measures discrimination ability across all classification thresholds.

Confusion matrices were plotted for all three models and ROC curves were overlaid on a single chart for direct comparison. A train vs. test accuracy gap analysis was performed to assess overfitting.

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4. Results and Discussion

4.1 Model Comparison

Table 4 presents the full performance comparison of all three models on the held-out test set (184 samples). Both Logistic Regression and Random Forest achieved the highest accuracy of 84.78%, followed by Decision Tree at 80.43%.

Table 4. Model Performance on Test Set

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.8478	0.8304	0.9118	0.8692	0.9063
Decision Tree	0.8043	0.8235	0.8235	0.8235	0.8815
Random Forest ★	**0.8478**	**0.8364**	0.9020	0.8679	**0.9168**

★ Random Forest achieved the highest ROC-AUC (0.9168), marking it as the overall best model.

4.2 Confusion Matrix Analysis (Best Model → Random Forest)

Table 5 presents the confusion matrix for the Random Forest model on the 184-sample test set.

Table 5. Confusion Matrix - Random Forest (Test Set, n=184)

	Predicted: No Disease	Predicted: Disease
Actual: No Disease	TN = 63	FP = 19
Actual: Disease	FN = 10	TP = 92

The Random Forest model correctly identified 92 of 102 disease-positive patients (Recall = 0.90) and 63 of 82 disease-negative patients. Only 10 disease cases were missed (false negatives), a critical metric in clinical screening.

4.3 Discussion

Best model and baseline comparison: Random Forest achieved the highest ROC-AUC (0.9168), confirming its superiority in discriminating between classes across all decision thresholds. It outperformed the Decision Tree baseline by 3.53 percentage points in accuracy and by 0.0353 in ROC-AUC. Logistic Regression, despite its simplicity, matched Random Forest in accuracy (84.78%) and achieved a very competitive ROC-AUC of 0.9063, demonstrating that a well-regularised linear model can be highly effective for this clinical dataset.

Precision vs. Recall trade-off: All three models show higher recall than precision for the heart disease class, which is the clinically desirable direction: it is more important to correctly identify patients with heart disease (minimising false negatives) than to be overly conservative. Logistic Regression achieved the highest recall (0.9118), meaning it missed only 9 of 102 positive cases.

Feature importance: The Random Forest feature importance scores reveal that the top five predictors are: exercise-induced angina (exang, 0.162), ST depression (oldpeak, 0.120), atypical chest pain (cp_atypical angina, 0.107), maximum heart rate achieved (thalch, 0.104), and patient age (0.085). All five have well-established clinical linkage to cardiovascular disease, which validates the model's learning and increases clinical trust.

Overfitting analysis: The train-test accuracy gap was minimal for Logistic Regression (−0.016, indicating slight underfitting), small for Decision Tree (0.019) and moderate for Random Forest (0.039). While Random Forest has the largest gap, the absolute difference remains under 4 percentage points, suggesting acceptable generalisation without severe overfitting, partially due to the depth constraint (max_depth = 5) found during tuning.

Comparison with related work: The results are broadly aligned with the literature. Anderies et al. [5] reported 85% accuracy for SVM on the Cleveland dataset. The current study achieves a comparable 84.78% with Logistic Regression and Random Forest on the larger multi-centre dataset, suggesting that performance is robust across centre sources. Mishra et al. [4] found that neural networks outperformed simpler models without PCA; this study demonstrates that Random Forest achieves strong performance without any dimensionality reduction.

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5. Conclusion

5.1 Summary

This study applied a complete machine learning pipeline to the Heart Disease UCI dataset (920 records, 4 clinical centres) to predict the binary presence or absence of cardiovascular disease. Three classification algorithms were compared after rigorous preprocessing, feature encoding, and hyperparameter tuning via 5-fold stratified cross-validation. Both Logistic Regression and Random Forest achieved 84.78% test accuracy, with Random Forest leading in ROC-AUC (0.9168) and Logistic Regression achieving the highest recall (0.9118). The most clinically meaningful predictors identified were exercise-induced angina, ST depression, chest pain type, maximum heart rate, and age. The results demonstrate that well-tuned standard ML models can reliably support early CVD screening, with potential to assist clinicians in prioritising high-risk patients.

5.2 Limitations

- Dataset size and population scope: Despite being the largest publicly available heart disease dataset, 920 records limits the statistical power for detecting rare subgroup effects. The dataset covers four specific clinical centres, which may introduce selection bias and limit generalisability to other populations.
- High missingness in key features: The ca (66.4%) and thal (52.8%) columns had very high proportions of missing values. Although KNN imputation was applied, imputed values inevitably introduce uncertainty and may not reflect the true clinical measurements. Sensitivity analysis comparing results with and without these features would strengthen confidence in conclusions.
- No external validation: All results are reported on the held-out portion of the same dataset. External validation on an independent clinical cohort, particularly one from a different geographic or demographic context, is necessary to assess true generalisability.
- Binary classification only: The model predicts only the presence or absence of CVD, not its severity. A multi-class model that predicts severity levels (as encoded in the original num variable, 0–4) would be more clinically actionable.

5.3 Future Work

- Advanced ensemble methods: Gradient boosting algorithms (XGBoost, LightGBM) and stacking ensembles combining all three models should be evaluated, as they have shown state-of-the-art performance on tabular clinical data in the literature.
- Extended feature engineering: Interaction terms between clinically correlated features (e.g., age × thalch, exang × oldpeak) and temporal features (e.g: time since first diagnosis) could improve predictive power.
- Handling class imbalance with SMOTE: Although the imbalance in this dataset is mild, applying SMOTE (Synthetic Minority Over-sampling Technique) or class-weight adjustments could further improve recall for the positive class.
- Clinical deployment study: A prospective study integrating the best model into a clinical decision support tool, with physician feedback and real-time performance monitoring, would be the logical next step toward real-world impact.

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